

NEW BEDFORD, MA, USA

WMO: 725065

Lat: 41.676N

Lon: 70.958W

Elev: 80

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 94726

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.5	(c) 12.1	(d) -8.8	(e) 3.4	(f) 12.2	(g) -4.2	(h) 4.4	(i) 16.5	(j) 26.8	(k) 37.2	(l) 24.7	(m) 36.9	(n) 7.6	(o) 290	(p) 0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	88.3	73.8	85.1	71.9	82.3	70.4	76.4	84.0	74.9	81.3	73.4	78.9	9.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	127.2	79.7	72.8	121.9	78.1	71.9	118.2	77.1	39.8	83.9	38.3	81.6	37.0	79.1	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.1	19.8	17.8	0.6	93.8	6.0	3.7	-3.8	96.5	-7.3	98.6	-10.7	100.7	-15.1	103.3	
			DB												
			WB	-0.6	79.0	5.7	1.8	-4.7	80.2	-8.1	81.2	-11.3	82.2	-15.4	83.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	30.1	32.1	37.8	47.6	57.1	65.8	72.3	70.9	63.9	53.2	43.4	35.6
	DBStd	16.42	9.90	8.66	8.42	7.10	6.95	6.48	5.35	5.43	6.59	7.58	8.18	8.81
	HDD50	2381	618	504	387	127	12	0	0	0	0	52	228	453
	HDD65	5747	1082	922	842	524	262	68	7	11	98	374	647	910
	CDD50	2721	2	2	10	55	232	474	690	648	418	150	32	8
	CDD65	610	0	0	0	1	18	92	232	194	65	8	0	0
	CDH74	4489	0	0	1	24	179	675	1787	1440	349	34	0	0
	CDH80	1198	0	0	0	6	43	171	551	372	51	4	0	0
(14) Wind	WSAvg	7.7	8.5	8.5	9.1	8.9	8.0	7.2	6.9	6.2	6.3	7.2	7.5	7.8
(15) Precipitation	PrecAvg	48.2	3.8	3.6	5.0	4.2	3.6	3.5	3.1	3.8	4.3	4.0	4.3	4.8
	PrecMax	58.8	6.6	7.8	12.6	8.1	7.3	10.7	8.0	7.7	10.6	10.9	7.0	9.0
	PrecMin	36.2	1.9	2.0	0.4	1.5	1.1	0.4	0.8	0.4	0.9	0.2	0.8	1.6
	PrecStd	6.0	1.3	1.4	2.8	1.7	1.4	2.6	1.7	1.7	2.3	2.4	1.7	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.1	57.6	67.9	76.9	84.9	89.5	92.8	90.9	85.3	78.8	66.5	62.1
		MCWB	55.6	51.5	55.7	61.8	69.7	73.4	75.7	75.0	74.3	67.9	60.9	58.1
	2%	DB	54.0	53.3	59.5	68.4	79.0	84.4	89.1	87.1	80.9	72.5	63.7	57.4
		MCWB	51.2	49.6	49.5	55.4	64.8	70.6	74.1	73.7	69.4	65.6	59.3	54.6
	5%	DB	49.6	49.4	54.4	63.8	73.8	81.3	85.7	83.9	78.4	69.3	61.0	54.4
		MCWB	46.3	45.0	48.1	53.4	61.9	69.1	72.4	71.8	68.6	63.4	56.6	51.8
	10%	DB	45.4	45.6	51.3	59.7	69.9	77.8	82.6	81.4	75.1	66.2	57.5	50.4
		MCWB	42.1	40.7	45.8	50.6	59.8	66.6	71.0	70.5	67.5	61.2	53.2	47.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.4	54.1	58.3	63.9	71.3	75.7	78.5	78.6	75.8	70.8	64.0	59.6
		MCDB	57.8	54.9	65.0	70.9	82.9	85.8	89.7	87.6	81.6	74.6	65.2	61.4
	2%	WB	52.1	50.5	53.3	59.2	67.4	73.3	76.5	76.6	73.2	67.4	60.9	55.5
		MCDB	53.3	52.6	56.1	66.0	75.3	81.7	85.2	83.0	77.2	70.6	63.0	57.5
	5%	WB	46.4	46.0	49.9	55.9	64.5	71.1	74.9	75.0	71.7	64.9	57.8	52.2
		MCDB	48.4	48.3	53.9	60.8	71.1	77.8	81.9	80.4	75.2	68.1	60.3	53.9
	10%	WB	42.0	41.4	46.0	53.0	61.8	69.2	73.3	73.5	69.9	62.3	54.3	47.6
		MCDB	45.1	45.2	49.6	57.3	67.4	74.9	79.3	78.0	73.6	65.4	57.0	49.6
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	16.6	17.0	18.3	18.5	18.0	18.3	18.6	19.1	18.9	17.4	15.9
		MCDBR	20.4	20.3	21.6	24.6	24.5	22.2	21.1	20.5	20.5	20.3	18.9	18.7
		MCWBR	18.9	16.7	14.8	14.1	12.8	11.0	9.1	9.4	11.7	13.7	15.8	17.8
		MCDBR	19.4	19.8	19.5	19.9	21.2	18.5	17.1	16.3	15.2	17.4	17.0	18.3
	5% WB	MCWBR	19.3	18.5	15.9	14.0	12.3	10.5	8.4	8.4	10.3	14.0	16.2	18.3
(40) Clear-Sky Solar Irradiance	taub		0.303	0.311	0.335	0.372	0.410	0.460	0.483	0.438	0.373	0.342	0.315	0.303
	taud		2.509	2.491	2.467	2.390	2.320	2.221	2.162	2.280	2.437	2.531	2.566	2.561
	Ebn at Noon		271	286	290	285	275	261	253	262	273	271	265	262
	Edh at Noon		23	27	31	36	40	44	47	40	32	26	22	20
(44) All-Sky Solar Radiation	RadAvg		577	858	1178	1519	1694	1836	1896	1704	1385	938	638	486
	RadStd		44	83	87	162	167	176	103	88	99	82	45	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+175	+85

Nomenclature: See separate page